DELETE OPERATIONS

MongoDB provides:

```
deleteOne()
deleteMany()
```

---

13. `deleteOne()`

`deleteOne()` deletes **one document** matching the condition.

Example

Delete student with roll number 105:

```
db.students.deleteOne({ rollno: 105 })
```

Output:

```
{
    acknowledged: true,
    deletedCount: 1
}
```

Check:

```
db.students.find({ rollno: 105 })
```

You should get no document.

---

14. Delete Using Name

```
db.students.deleteOne({ name: "Amit" })
```

This deletes one matching document.

Important warning for students

If multiple documents have the same name, `deleteOne()` deletes only **one matching document**.

For safer deletion, use a unique field such as:

```
rollno
```

Example:

```
db.students.deleteOne({ rollno: 103 })
```

---

15. `deleteMany()`

`deleteMany()` deletes **all documents matching the condition**.

Example

Delete all Semester 3 students:

```
db.students.deleteMany({ semester: 3 })
```

Output might be:

```
{
    acknowledged: true,
    deletedCount: 2
}
```

Now:

```
db.students.find()
```

The Semester 3 students will be gone.

---

16. Delete Students Based on Marks

Delete students whose marks are less than 60:

```
db.students.deleteMany({
    marks: { $lt: 60 }
})
```

Here:

```
$lt = less than
```

---

17. Delete Students from a Particular City

For example:

```
db.students.deleteMany({
    city: "Anjar"
```

```
})
```

This deletes **all documents** where city is Anjar.

---

18. Delete All Documents from Collection

⚠️ **Important classroom warning**

If you want to remove every document from `students`:

```
db.students.deleteMany({})
```

The empty filter:

```
{}
```

means:

Match every document.

So:

```
db.students.deleteMany({})
```

means:

Delete every document from the `students` collection.

The collection itself remains.

---

19. Delete Collection

This is different.

If you want to delete the **entire collection**:

```
db.students.drop()
```

Output:

```
true
```

Now:

```
show collections
```

The `students` collection will no longer exist.

Difference

```
db.students.deleteMany({})
```

➡️ deletes all documents

```
db.students.drop()
```

➡️ deletes the entire collection

---

20. Delete Database

If you want to delete the entire current database:

```
db.dropDatabase()
```

For example:

```
use college
db.dropDatabase()
```

Output:

```
{ ok: 1, dropped: 'college' }
```

⚠️ **Use this only as a demonstration.**

⭐ Very Important Difference for Students

Command	What it does
`updateOne()`	Updates one matching document
`updateMany()`	Updates all matching documents
`replaceOne()`	Replaces one complete document
`$set`	Changes/adds a field
`$inc`	Increases/decreases numeric value
`$unset`	Removes a field
`$rename`	Renames a field
`deleteOne()`	Deletes one matching document
`deleteMany()`	Deletes all matching documents
`drop()`	Deletes entire collection

Command	What it does
`dropDatabase()`	Deletes entire database

🎓 Live Classroom Practical

I recommend you perform this sequence **exactly in front of your students**.

1. Select database

```
use college
```

2. Insert data

```
db.students.insertMany([
    {rollno:101, name:"Rahul", course:"BCA", semester:5, marks:75,
city:"Anjar"},
    {rollno:102, name:"Priya", course:"BCA", semester:3, marks:82,
city:"Gandhidham"},
    {rollno:103, name:"Amit", course:"BCA", semester:5, marks:68,
city:"Bhuj"},
    {rollno:104, name:"Neha", course:"BCA", semester:3, marks:91,
city:"Anjar"},
    {rollno:105, name:"Karan", course:"BCA", semester:6, marks:59,
city:"Mandvi"}
])
```

3. Display

```
db.students.find().pretty()
```

4. Update Rahul's marks

```
db.students.updateOne(
    {rollno:101},
    {$set:{marks:85}}
)
```

5. Check

```
db.students.findOne({rollno:101})
```

6. Increase all Semester 5 marks by 5

```
db.students.updateMany(
    {semester:5},
    {$inc:{marks:5}}
)
```

7. Check

```
db.students.find({semester:5})
```

8. Add mobile number

```
db.students.updateOne(
    {rollno:101},
    {$set:{mobile:"9876543210"}}
)
```

9. Remove mobile number

```
db.students.updateOne(
    {rollno:101},
    {$unset:{mobile:""}}
)
```

10. Delete one student

```
db.students.deleteOne({rollno:105})
```

11. Delete students with marks below 70

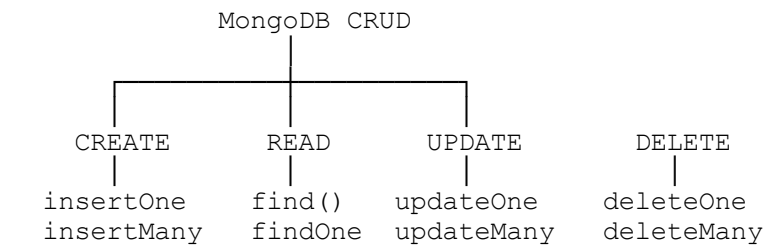
```
db.students.deleteMany({
    marks:{$lt:70}
})
```

12. Final display

```
db.students.find().pretty()
```

🧠 Simple Explanation to Give Students

You can explain CRUD like this:



UPDATE

Tell students:

"Update means changing existing data without deleting the complete document."

Example:

```
db.students.updateOne(
    {rollno:101},
    {$set:{marks:90}}
)
```

DELETE

Tell students:

"Delete means removing an existing document from a collection."

Example:

```
db.students.deleteOne({rollno:101})
```

⚠️ 5 Important Warnings for Students

1. Be careful with `{}`

```
db.students.deleteMany({})
```

means **delete everything in that collection**.

2. `updateOne()` doesn't mean "update the first record"

It means:

Update **one matching document**.

3. `updateMany()` can change many records

Always run:

```
db.students.find({condition})
```

before running `updateMany()` or `deleteMany()`.

For example:

```
db.students.find({semester:5})
```

Then:

```
db.students.updateMany(
    {semester:5},
```

```
    {$inc:{marks:5}}
)
```

4. `deleteMany()` is permanent

There is no simple "Ctrl+Z" in MongoDB.

5. Prefer unique conditions for deletion

Better:

```
db.students.deleteOne({rollno:101})
```

than:

```
db.students.deleteOne({name:"Rahul"})
```

because `rollno` should uniquely identify a student.

📝 Student Practical Exercises

After demonstrating the above, give students these tasks:

Exercise 1

Change Priya's city to `Anjar`.

```
Expected concept: updateOne() + $set
```

Exercise 2

Increase all BCA students' marks by 3.

```
Expected concept: updateMany() + $inc
```

Exercise 3

Add an `email` field to Rahul.

```
Expected concept: $set
```

Exercise 4

Remove Rahul's email field.

```
Expected concept: $unset
```

Exercise 5

Rename `city` to `address` for one student.

`Expected concept: $rename`

Exercise 6

Delete student with roll number 103.

`Expected concept: deleteOne()`

Exercise 7

Delete all students whose marks are below 60.

`Expected concept: deleteMany() + $lt`

Exercise 8

Delete all Semester 3 students.

`Expected concept: deleteMany()`

Exercise 9

Display the remaining students.

```
db.students.find().pretty()
```